



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

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## SAFETY DATA SHEET

### Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

### 1. CHEMICAL IDENTIFICATION

#### Product Identification

Trade name: Kikae

Product name: fomesafen 25% W/V SL

Product use: herbicide

#### Relevant identified uses of the substance/mixture and uses advised against:

Used as selective herbicide offering pre- and post-emergence management of difficult weeds.

### 2. HAZARD IDENTIFICATION

Classification of risk: WHO III; EPA II

#### Potential Health Effects

Causes eye and skin irritation. It is toxicity if swallowed or absorbed through skin. Irritate to eyes and skin. May causes skin sensitization if contact skin.

#### Hazardous Decomposition Products

Can decompose at high temperatures forming toxic gases.

#### Unusual fire, explosion and reactivity hazards

During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

### 3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: 5-(2-Chloro-4-(trifluoromethyl)phenoxy)-N-(methylsulfonyl)-2-nitrobenzamide

Common name: fomesafen

CAS Number: 72178-02-0

Formula: C<sub>15</sub>H<sub>10</sub>ClF<sub>3</sub>N<sub>2</sub>O<sub>6</sub>S

Molecular weight: 438.76 g/mol



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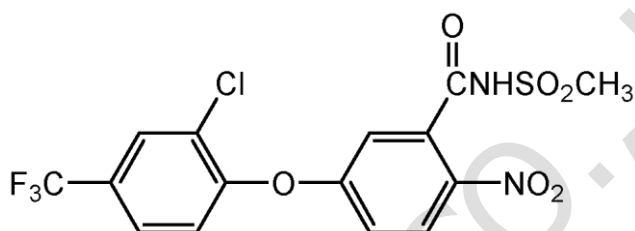
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Ingredients and composition:

| Component          | % Content  |
|--------------------|------------|
| Active ingredient  |            |
| - Fomesafen        | 25.0 min   |
| 30% liquid caustic | 7.0 max    |
| Surfactant         |            |
| - CL-18B           | 5.0 max    |
| Water              | Up to 100% |

Structural formula:



#### 4. FIRST AID MEASURES

Have the product container, label or Safety Data Sheet with you when calling a Poison Control Center or doctor, or going for treatment.

##### Description of first aid measures

In case of swallowed:

Call a Poison Control Center or doctor immediately for treatment advice.

Have the person sip a glass of water if able to swallow.

Do not induce vomiting unless told to do so after calling a poison control center or doctor.

Do not give anything by mouth to an unconscious person.

In case of eyes contact:

Hold eye open and rinse slowly and gently with water for 15-20 minutes.

Remove contact lenses, if present, after 5 minutes, then continue rinsing.

Call a Poison Control Center or doctor for treatment advice.

In case of skin contact:

Take off contaminated clothing.

Rinse skin immediately with plenty of water for 15-20 minutes.

Call a Poison Control Center or doctor for treatment advice.

In case of Inhaled contact:

Move person to fresh air. If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible.

Call a Poison Control Center or doctor for treatment advice.



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Note to Physician:

Probable mucosal damage may contraindicate gastric lavage.

## 5. FIRE FIGHTING MEASURES

Fire and Explosion:

Flash point: > 212 °F

Unusual Fire, Explosion and Reactivity Hazards:

During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

Fire Fighting Equipment:

Use dry chemical, foam or CO<sub>2</sub> extinguishing media. Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area, and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

## 6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

In case of leak or spill:

- (1) Control the spill at its source.
- (2) Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems or anybody of water.
- (3) Clean up spills immediately, observing precautions outlined in **Section 8**.
- (4) Cover entire spill with absorbing material and place into compatible disposal container.
- (5)
- (6) Disposal of the absorbed material will depend upon the extent of the spill.
  - For quantities up to 50L of product bury in a secure landfill site.
  - For quantities greater than 50L seek advice from the manufacturer (use emergency contact number below) before attempting disposal. Contain in a secure location until disposal method is established.
- (7) Decontaminate the spill area with detergent and water and rinse with the smallest volume of water practicable

## 7. HANDLING AND STORAGE

Store the material in a well-ventilated, secure area out of reach of children and domestic animals.

Do not store, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco



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use, and cosmetic application in areas where there is a potential for exposure to the material.

Wash thoroughly with soap and water after handling.

## 8. EXPOSURE CONTROL/PERSONAL PROTECTION

### Personal Protective Equipment

#### Ingestion:

Prevent eating, drinking, tobacco usage and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

#### Eyes Protection:

Where eye contact is likely, use chemical splash goggles. Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

#### Skin Protection:

Where contact is likely, wear chemical-resistant (such as nitrile or butyl) gloves, coveralls, socks and chemical-resistant footwear. For overhead exposure, wear chemical-resistant headgear.

#### Inhalation Protection:

Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below exposure limits. A NIOSH-certified combination air-purifying respirator with an N, P or R 95 or HE class filter and an organic vapor cartridge may be permissible under certain circumstances where airborne concentrations are expected to exceed exposure limits. Protection provided by air-purifying respirators is limited. Use a pressure demand atmosphere-supplying respirator if there is any potential for uncontrolled release, exposure levels are not known, or under any other circumstances where air-purifying respirators may not provide adequate protection.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                   |                                |
|-------------------|--------------------------------|
| Physical stage:   | Liquid (microemulsion)         |
| Odor:             | Not determined                 |
| pH:               | 6.0 – 9.0                      |
| Boiling point:    | 100.6 °C – hydrocarbon solvent |
| Specific gravity: | 1.125 g/ml                     |

## 10. STABILITY AND REACTIVITY

|                           |                                                 |
|---------------------------|-------------------------------------------------|
| Stability:                | Stable under normal use and storage conditions. |
| Hazardous Polymerization: | Will not occur.                                 |



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|                                   |                                                         |
|-----------------------------------|---------------------------------------------------------|
| Conditions to Avoid:              | None known.                                             |
| Materials to Avoid:               | Oxidizing agents.                                       |
| Hazardous Decomposition Products: | Can decompose at high temperatures forming toxic gases. |

### 11. TOXICOLOGICAL INFORMATION

#### Information on Acute toxicity for end product, Fomesafen:

|                             |                                      |
|-----------------------------|--------------------------------------|
| LD50 (Oral, rat):           | > 3,863 mg/kg bw SLIGHTLY TOXIC      |
| LD50 (Dermal, rat):         | > 2,000 mg/kg bw SLIGHTLY TOXIC      |
| LC50 (Inhalation, rat, 4h): | > 2.1 mg/L air PRACTICALLY NON-TOXIC |
| Skin irritation (rabbit):   | MODERATELY irritating.               |
| Eye irritation (rabbit):    | MODERATELY irritating.               |
| Skin sensitization:         | MODERATE skin sensitizer             |

#### Carcinogenicity data

Increased rates of malignant liver tumors in a 2-yr mouse feeding study (1000 ppm), but the results are not considered relevant to man (base on Fomesafen technical).

#### Reproductive/Development Effects-Fomesafen sodium salt

Non-genotoxic in vitro and in vivo assays. Hepatic changes and a small effect on reproductive performance seen in a 3-generation rat study at the 1000 ppm dose level (based on Fomesafen technical).

#### Toxicity of other components

May cause severe eye irritation, possible burns. Causes skin irritation. Ingestion causes gastrointestinal irritation. May cause central nervous system depression, characterized by excitement, followed by headache, dizziness, drowsiness and nausea. Inhalation of high concentrations may cause central nervous system effects.

#### Target Organs

|                                             |                                         |
|---------------------------------------------|-----------------------------------------|
| Active ingredients (fomesafen sodium salt): | Liver (based on fomesafen technical)    |
| Inert Ingredients (Cyclohexanol)            | Eyes, skin, gastrointestinal tract, CNS |

### 12. ECOLOGICAL INFORMATION

#### Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to Fomesafen technical:

#### **Fish – Acute toxicity:**

TOXIC to fish (fomesafen sodium salt)



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Rainbow trout,

LC50 (96 h): 170 mg/L

Bluegill sunfish,

LC50 (96 h): 1,507 mg/L

Daphnia magna,

EC50 (48 h): 0.33 g/L

### **Birds – Acute toxicity:**

Bobwhite quail,

LC50 (5 days): > 20,000 mg/kg

Mallard ducks,

LD50: > 5,000 mg/kg

LC50 (5 days): > 20,000 mg/kg dietary

### **Algae – Acute toxicity:**

Green algae,

EC50: 170 µg/L

Eco-Chronic Toxicity

Fomesafen sodium salt: Not available

Environmental fate: the information presented here is for the active ingredient, fomesafen. Persistent in soil. Stable in water. Highly mobile in soil. Will leach. Sinks in water

## **13. DISPOSAL CONSIDERATIONS**

Dispose of empty, used containers by:

Do not reuse product containers. Dispose of product containers, waste containers, and residues according to local, state, and federal health and environmental regulations.

Characteristic waste:

Not applicable

Listed waste:

Not applicable

## **14. TRANSPORT INFORMATION**

Not classified as Dangerous for transport

## **15. REGULATORY INFORMATION**

Not applicable



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## 16. OTHER INFORMATION

All information and instructions provided in this Safety Data Sheet (SDS) are based on the current state of scientific and technical knowledge at the date indicated on the present SDS and are presented in good faith and believed to be correct. This information applies to the product as such. In case of new formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons on receipt of this SDS to ensure that the information contained herein is properly read and understood by all people who may use, handle, dispose or in any way come in contact with the product. If the recipients subsequently produce formulations containing this product, it is recipient sole responsibility to ensure the transfer of all relevant information from this SDS to their own SDS.